

Common Lung Disorder

Pneumonia

1. Pneumonia is defined as an acute respiratory illness associated with recently developed radiological pulmonary shadowing which may be segmental, lobar or multilobar.
2. Pneumonias are usually classified as community or hospital acquired, or those occurring in immune compromised hosts.
3. Lobar pneumonia is a radiological and pathological term referring to homogeneous consolidation of one or more lung lobes, often with associated pleural inflammation
4. Bronchopneumonia refers to more patchy alveolar consolidation associated with bronchial and bronchiolar inflammation often affecting both lower lobes.

Community-acquired pneumonia (CAP)

1. It is one of the lower respiratory tract infections.
2. The incidence varies with age, being much higher in the very young and very old, in whom the mortality rates are also much higher.
3. World-wide, CAP continues to kill more children than any other illness.
4. Most cases are spread by droplet infection and occur in previously healthy individuals but several factors may impair the effectiveness of local defences and predispose to CAP.
5. *Strep. Pneumonia* remains the most common infecting agent, and thereafter, the likelihood that other organisms may be involved depends on the age of the patient and the clinical context.
6. Viral infections are an important cause of CAP in children, and their contribution to adult CAP is increasingly recognised.

Factors that predispose to pneumonia

- Cigarette smoking
- Old age

- Upper respiratory tract
- Recent influenza infection

Infections

- Pre-existing lung disease
- Alcohol
- HIV
- Corticosteroid therapy
- Indoor air pollution

Clinical features

1. Fever
2. Rigor
3. Shivering
4. Vomiting predominate
5. The appetite is usually lost
6. Loss of appetite is usually present
7. Headache is common.
8. Pulmonary symptoms include
 - Breathlessness and cough, which at first is characteristically short, painful and dry, but later accompanied by the expectoration of mucopurulent sputum
 - Rust-coloured sputum may be seen in patients with *Strep. pneumoniae* infection,
 - Occasional patient may report haemoptysis.
 - Pleuritic chest pain may be a presenting feature and on occasion may be referred to the shoulder or anterior abdominal wall.

Physical examination findings

1. Upper abdominal tenderness is sometimes apparent in patients with lower lobe pneumonia or if there is associated hepatitis.
2. Proteinaceous fluid and inflammatory cells congest the airspaces, leading to consolidation of lung tissue
3. This improves the conductivity of sound to the chest wall and the clinician may hear

bronchial breathing and whispering pectoriloquy

4. Crackles are often also detected.

Investigations

X-ray :

1. A chest X-ray usually provides confirmation of the diagnosis.
2. In lobar pneumonia, a homogeneous opacity localised to the affected lobe or segment usually appears within 12-18 hours of the onset of the illness.
3. Radiological examination is helpful if a complication such as parapneumonic effusion, intrapulmonary abscess formation or empyema is suspected.

Microbiological tests

1. **Sputum:** direct smear by Gram and Ziehl-Neelsen stains.
2. **Culture** and antimicrobial sensitivity testing
3. **Blood culture:** frequently positive in pneumococcal pneumonia
4. **Serology:** acute and convalescent titres for *Mycoplasma*, *Chlamydia*, *Legionella*, and viral infections.
5. **Pneumococcal** antigen detection in serum or urine
6. **PCR:** *Mycoplasma* can be detected from swab of oropharynx. Severe community-acquired pneumonia

The above tests plus consider :

7. **Tracheal aspirate,** induced sputum, bronchoalveolar lavage, protected brush specimen or percutaneous needle aspiration. Direct fluorescent antibody stain for *Legionella* and viruses
8. **Serology:** *Legionella* antigen in urine. Pneumococcal antigen in sputum and blood. Immediate IgM for *Mycoplasma*
9. **Cold agglutinins:** positive in 50% of patients with *Mycoplasma*
10. A full range of microbiological tests should be performed on patients with severe CAP

11. The identification of *Legionella pneumophila* has important public health implications and requires notification.

12. In patients who do not respond to initial therapy, microbiological results may allow its appropriate modification.

13. Microbiology also provides useful epidemiological information.

Pulse oximetry

1. Pulse oximetry provides a non-invasive method of measuring arterial oxygen saturation (SaO₂) and monitoring response to oxygen therapy SpO₂.

Arterial blood gas

1. An arterial blood gas is important in those with SaO₂ < 93% or with features of severe pneumonia, to identify ventilatory failure or acidosis.

Full blood count

1. The white cell count may be normal or only marginally raised in pneumonia caused by atypical organisms, whereas a neutrophil leucocytosis of more than $15 \times 10^9/L$ favours a bacterial aetiology.
2. A very high ($> 20 \times 10^9/L$) or low ($< 4 \times 10^9/L$) white cell count may be seen in severe pneumonia.

Biochemical tests

1. Urea and electrolytes and liver function tests should also be checked.

Acute phase reactants

1. The C-reactive protein (CRP) is typically elevated.

Differential diagnosis of pneumonia

- Pulmonary infarction
- Pulmonary/pleural TB
- Pulmonary oedema (can be unilateral)
- Pulmonary eosinophilia
- Malignancy: bronchoalveolar cell carcinoma
- Rare disorders: cryptogenic organising

pneumonia/bronchiolitis obliterans
organising pneumonia (COP/BOOP)

Chronic Obstructive Pulmonary Disease (COPD)

Chronic obstructive pulmonary disease (COPD) is defined as a disease state characterized by airflow limitation that is not fully reversible. COPD includes emphysema, an anatomically defined condition characterized by destruction and enlargement of the lung alveoli; chronic bronchitis, a clinically defined condition with chronic cough and phlegm; and small airways disease, a condition in which small bronchioles are narrowed. COPD is present only if chronic airflow obstruction occurs; chronic bronchitis without chronic airflow obstruction is not included within COPD. It is the fourth leading cause of death and affects >10 million persons in the United States. COPD is also a disease of increasing public health importance around the world. Estimates suggest that COPD will rise from the sixth to the third most common cause of death worldwide by 2020.

Risk Factors

Cigarette Smoking; Airway Responsiveness; Respiratory Infections; Occupational Exposures; Ambient Air Pollution; Passive, or Second-Hand, Smoking Exposure; and Genetic Considerations.

Pathophysiology

Pathologic changes in chronic obstructive pulmonary disease (COPD) occur in the large (central) airways, the small (peripheral) bronchioles, and the lung parenchyma. Most cases of COPD are the result of exposure to noxious stimuli, most often cigarette smoke. The normal inflammatory response is amplified in persons prone to COPD development. The pathogenic mechanisms are not clear but are most likely diverse. Increased numbers of activated polymorphonuclear leukocytes and macrophages release elastases in a manner that cannot be counteracted effectively by antiproteases, resulting in lung destruction.

Clinical Features

Patients typically present with a combination of signs and symptoms of chronic bronchitis, emphysema, and reactive airway disease. Symptoms include the following: Cough, usually worse in the mornings and productive of a small amount of colorless sputum; Acute chest illness; Breathlessness: The most significant symptom, but usually does not occur until the sixth decade of life; Wheezing: May occur in some patients, particularly during exertion and exacerbations. The sensitivity of physical examination in detecting mild to moderate COPD is relatively poor, but physical signs are quite specific and sensitive for severe disease.

Physical Examination

In the early stages of COPD, patients usually have an entirely normal physical examination. Current smokers may have signs of active smoking, including an odor of smoke or nicotine staining of fingernails. Patients with severe airflow obstruction may also exhibit use of accessory muscles of respiration, sitting in the characteristic "tripod" position to facilitate the actions of the sternocleidomastoid, scalene, and intercostal muscles. Patients may develop cyanosis, visible in the lips and nail beds.

Advanced disease may be accompanied by systemic wasting, with significant weight loss, bitemporal wasting, and diffuse loss of subcutaneous adipose tissue. This syndrome has been associated with both inadequate oral intake and elevated levels of inflammatory cytokines. Such wasting is an independent poor prognostic factor in COPD. Some patients with advanced disease have paradoxical inward movement of the rib cage with inspiration (Hoover's sign), the result of alteration of the vector of diaphragmatic contraction on the rib cage as a result of chronic hyperinflation.

Signs of overt right heart failure, termed *cor pulmonale*, are relatively infrequent since the advent of supplemental oxygen therapy.

Clubbing of the digits is not a sign of COPD,

and its presence should alert the clinician to initiate an investigation for causes of clubbing. In this population, the development of lung cancer is the most likely explanation for newly developed clubbing.

Findings in severe disease include the following: Tachypnea and respiratory distress with simple activities; Use of accessory respiratory muscles and paradoxical indrawing of lower intercostal spaces (Hoover sign); Cyanosis; Elevated jugular venous pulse (JVP).

Peripheral edema and Thoracic examination reveals the following: Hyperinflation (barrel chest); Wheezing - Frequently heard on forced and unforced expiration; Diffusely decreased breath sounds; Hyperresonance on percussion; Prolonged expiration; Coarse crackles beginning with inspiration in some cases

Certain characteristics allow differentiation between disease that is predominantly chronic bronchitis and that which is predominantly emphysema.

Chronic bronchitis characteristics include the following: Patients may be obese; frequent cough and expectoration are typical; Use of accessory muscles of respiration is common; coarse rhonchi and wheezing may be heard on auscultation and Patients may have signs of right heart failure (ie, cor pulmonale), such as edema and cyanosis.

Emphysema characteristics include the following: Patients may be very thin with a barrel chest; Patients typically have little or no cough or expectoration; Breathing may be assisted by pursed lips and use of accessory respiratory muscles; patients may adopt the tripod sitting position; The chest may be hyperresonant, and wheezing may be heard; Heart sounds are very distant.

Ancillary Investigations

Spirometry : The formal diagnosis of COPD is made with spirometry; when the ratio of forced expiratory volume in 1 second over forced vital capacity (FEV1/FVC) is less than 70% of that predicted for a matched control; it is diagnostic for a significant obstructive defect.

Criteria for assessing the severity of airflow obstruction (based on the percent predicted post bronchodilator FEV1) are as follows:

- Stage I (mild): FEV1 80% or greater of predicted
- Stage II (moderate): FEV1 50-79% of predicted
- Stage III (severe): FEV1 30-49% of predicted
- Stage IV (very severe): FEV1 less than 30% of predicted or FEV1 less than 50% and chronic respiratory failure

Arterial blood gas (ABG) findings are as follows :

- ABGs provide the best clues as to acuteness and severity of disease exacerbation
- Patients with mild COPD have mild to moderate hypoxemia without hypercapnia
- As the disease progresses, hypoxemia worsens and hypercapnia may develop, with the latter commonly being observed as the FEV1 falls below 1 L/s or 30% of the predicted value
- pH usually is near normal; a pH below 7.3 generally indicates acute respiratory compromise
- Chronic respiratory acidosis leads to compensatory metabolic alkalosis

In patients with emphysema, frontal and lateral chest radiographs reveal the following:

- Flattening of the diaphragm; Increased retrosternal air space; A long, narrow heart shadow and Rapidly tapering vascular shadows accompanied by hyperlucency of the lungs
- Radiographs in patients with chronic bronchitis show increased bronchovascular markings and cardiomegaly

Advantages of high-resolution CT include the following:

- Greater sensitivity than standard chest radiography
- High specificity for diagnosing emphysema (outlined bullae are not always visible on a radiograph)
- May provide an adjunctive means of

diagnosing various forms of COPD (eg, lower lobe disease may suggest alpha1-antitrypsin (AAT) deficiency)

Other tests are as follows:

- Hematocrit - Patients with polycythemia (hematocrit greater than 52% in men or 47% in women) should be evaluated for hypoxemia at rest, with exertion, or during sleep
- Serum potassium - Diuretics, beta-adrenergic agonists, and theophylline act to lower potassium levels
- Measure AAT in all patients younger than 40 years or in those with a family history of emphysema at an early age
- Sputum evaluation will show a transformation from mucoid in stable chronic bronchitis to purulent in acute exacerbations
- Pulse oximetry, combined with clinical observation, provides instant feedback on a patient's status
- Electrocardiography can help establish that hypoxia is not resulting in cardiac ischemia and that the underlying cause of respiratory difficulty is not cardiac in nature
- The distance walked in 6 minutes (6MWD) is a good predictor of all-cause and respiratory mortality in patients with moderate COPD; patients with COPD who desaturate during the 6MWD have a higher mortality rate than do those who do not desaturate
- Two-dimensional echocardiography can screen for pulmonary hypertension
- Right-sided heart catheterization can confirm pulmonary artery hypertension and gauge the response to vasodilators

Source - Harrison's Principles of Internal Medicine, 18th Edition and e medicine articles

Pleural effusion

The accumulation of serous fluid within the pleural space is termed pleural effusion.

Empyema

The accumulation of frank pus within the pleural space is termed empyema

Haemothorax

The accumulation of blood within the pleural space is termed haemothorax.

Chylothorax

The accumulation of chyle within the pleural space is termed chylothorax.

Causes of pleural effusion

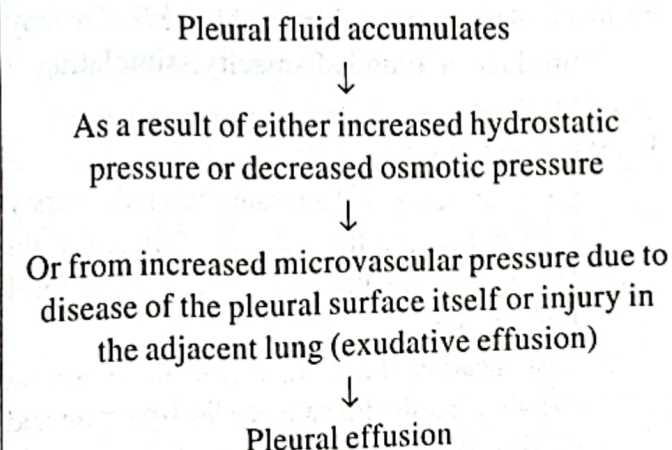
Common causes :

- Pneumonia (para-pneumonic effusion)
- Cardiac failure
- Subdiaphragmatic disorders (subphrenic abscess, pancreatitis etc.)
- Tuberculosis
- Pulmonary infarction
- Malignant disease

Uncommon causes :

- Hypoproteinaemia (nephrotic syndrome, liver failure, malnutrition)
- Meigs syndrome (ovarian tumour plus pleural effusion)
- Connective tissue diseases (particularly systemic lupus erythematosus (SLE) and rheumatoid arthritis)
- Myxoedema
- Uraemia
- Asbestos-related benign pleural effusion
- Acute rheumatic fever
- Post-myocardial infarction syndrome

Pathogenesis



Clinical assessment

Pain on inspiration and coughing

Signs

1. Tachypnoea
2. Absent breath sounds and vocal resonance
3. Stony dullness on percussion
4. Crackles above effusion
5. Pleurisy (a pleural rub) often precedes the development of an effusion, especially in patients with underlying pneumonia, pulmonary infarction or connective tissue disease.
6. Breathlessness is the only symptom related to the effusion itself, and its severity depends on the size and rate of accumulation.

Investigations**Imaging :**

1. The classical appearance of pleural fluid on the erect PA chest film is of a curved shadow at the lung base, blunting the costophrenic angle and ascending towards the axilla
2. Around 200 ml of fluid is required to be detectable on a PA chest X-ray
3. The smaller effusions can be identified by ultrasound or CT.
4. Previous scarring or adhesions in the pleural space can cause localised effusions.
5. Pleural fluid localised below the lower lobe ('subpulmonary effusion') simulates an elevated hemidiaphragm.
6. Fluid localised within an oblique fissure may produce a rounded opacity, simulating a tumour.

Ultrasonography :

7. It is more accurate than plain chest radiography for determining the volume of pleural fluid and frequently provides additional helpful information.
8. Visualisation of fluid facilitates skin marking to indicate a site for safe needle aspiration and guides pleural biopsy, increasing diagnostic yield.

9. The presence of loculation may suggest an evolving empyema or resolving haemothorax.
10. The technique may also distinguish pleural fluid from pleural thickening.

CT :

11. CT displays pleural abnormalities more readily than either plain radiography or ultrasound, and may distinguish benign from malignant pleural disease.

Pleural aspiration and biopsy :

12. In some clinical settings (e.g. left ventricular failure) it should not be necessary to sample fluid unless atypical features present.
13. Simple aspiration provides information on the colour and texture of fluid and on appearance alone may immediately suggest an empyema or chylothorax.
14. The presence of blood is consistent with pulmonary infarction or malignancy, but may represent a traumatic tap.

Biochemical analysis :

15. Biochemical analysis allows differentiation of transudates from exudates and Gram stain may indicate pneumonic effusion.

Cytological examination :

16. Cytological examination is essential, as the predominant cell type provides useful information.
17. A low pH suggests infection but may also be seen in rheumatoid arthritis, ruptured oesophagus or advanced malignancy.

Combining pleural aspiration with biopsy :

18. Combining pleural aspiration with biopsy increase the diagnostic yield, particularly when guided by either ultrasound or CT.

Thoracoscopy :

19. The best results are obtained from video-assisted thoracoscopy, allowing the operator to visualise the pleura and guide the biopsy directly

Light's criteria for distinguishing pleural transudate from exudate

Pleural fluid is an exudate if one more of the following criteria are met.

- Pleural fluid protein:serum protein ratio > 0.5
- Pleural fluid LDH:serum LDH ratio > 0.6
- Pleural fluid LDH > two-thirds of the upper limit of normal serum LDH
(LDH = lactate dehydrogenase)

Bronchitis

Bronchitis is characterized by inflammation of the bronchial tubes (bronchi), the air passages that extend from the trachea into the small airways and alveoli.

Etiology

Respiratory viruses are the most common causes of acute bronchitis, and cigarette smoking is indisputably the predominant cause of chronic bronchitis.

Viral and bacterial infections in acute bronchitis

The most common viruses include influenza A and B, parainfluenza, respiratory syncytial virus, and coronavirus, although an etiologic agent is identified only in a minority of cases.

Smoking and other causes of chronic bronchitis

Cigarette smoking is indisputably the predominant cause of chronic bronchitis. Common risk factors for acute exacerbations of chronic bronchitis are advanced age and low forced expiratory volume in one second (FEV₁). Most (70-80%) acute exacerbations of chronic bronchitis are estimated to be due to respiratory infections.

Clinical Features

A complete history must be obtained, including information on exposure to toxic substances and smoking. Symptoms of bronchitis include the following: Cough (the most commonly observed symptom); Sputum production (clear, yellow,

green, or even blood-tinged); Fever (relatively unusual; in conjunction with cough, suggestive of influenza or pneumonia); Nausea, vomiting, and diarrhoea (rare); General malaise and chest pain (in severe cases); Dyspnea and cyanosis (only seen with underlying chronic obstructive pulmonary disease [COPD] or another condition that impairs lung function); Sore throat; Runny or stuffy nose; Headache; Muscle aches and Extreme fatigue

Physical Examination Findings

Diffuse wheezes, high-pitched continuous sounds, and the use of accessory muscles (in severe cases). Diffuse diminution of air intake or inspiratory stridor (indicative of bronchial or tracheal obstruction). Sustained heave along the left sternal border (indicative of right ventricular hypertrophy secondary to chronic bronchitis). Clubbing on the digits and peripheral cyanosis (indicative of cystic fibrosis). Bullous myringitis (suggestive of mycoplasmal pneumonia). Conjunctivitis, adenopathy, and rhinorrhea (suggestive of adenoviral infection)

Ancillary Investigations

Bronchitis may be suspected in patients with an acute respiratory infection with cough; yet, because many more serious diseases of the lower respiratory tract cause cough, bronchitis must be considered a diagnosis of exclusion.

Complete blood count (CBC) with differential-

Procalcitonin levels (to distinguish bacterial from nonbacterial infections)

Sputum cytology (if the cough is persistent)

Blood culture (if bacterial super infection is suspected). Chest radiography (if the patient is elderly or physical findings suggest pneumonia). Bronchoscopy (to exclude foreign body aspiration, tuberculosis, tumors, and other chronic diseases)

Influenza tests

Spirometry

Laryngoscopy (to exclude epiglottitis)

Bronchiectasis

1. Bronchiectasis means abnormal dilatation of the bronchi.
2. Chronic suppurative airway infection with sputum production, progressive scarring and lung damage are present, whatever the cause.

Aetiology and pathogenesis

A. Congenital

1. Cystic fibrosis
2. Ciliary dysfunction syndromes
Primary ciliary dyskinesia (immotile cilia syndrome)
Kartagener's syndrome sinusitis and transposition of the Primary hypogammaglobulinaemia

B. Acquired: children

1. Pneumonia (complicating whooping cough or measles)
2. Primary TB
3. Inhaled foreign body

C. Acquired: adults

1. Suppurative pneumonia
2. Pulmonary TB
3. Allergic bronchopulmonary aspergillosis complicating asthma
4. Bronchial tumours

Pathogenesis

Bronchiectasis may result from a congenital defect affecting airway ion transport or ciliary function, such as cystic fibrosis



Acquired secondary to damage to the airways by a destructive infection, inhaled toxin or foreign body



The result is chronic inflammation and infection in airways



Chronic inflammatory and fibrotic changes are usually found in the surrounding lung tissue, resulting in progressive destruction of the normal lung architecture in advanced cases.



Localised bronchiectasis may occur



Due to the accumulation of pus beyond an obstructing bronchial lesion



Such as enlarged tuberculous hilar lymph nodes, a bronchial tumour or an inhaled foreign body (e.g. an aspirated peanut).

Clinical features

Symptoms of bronchiectasis :

1. Cough-Chronic productive cough due to accumulation of pus in dilated bronchi; usually worse in mornings and often brought on by changes of posture.
2. Sputum often copious and persistently purulent in advanced disease.
3. Halitosis is a common accompanying feature
4. Pneumonia and pleurisy occurs
5. Haemoptysis- Can be slight or massive and is often recurrent. Usually associated with purulent sputum or an increase in sputum purulence. Can, however, be the only symptom in so-called 'dry bronchiectasis'
6. Poor general health-When disease is extensive and sputum persistently purulent, there may be associated weight loss, anorexia, lassitude, low-grade fever, and failure to thrive in children.
7. In these patients, digital clubbing is common

Physical signs

1. When there are large amounts of sputum in the bronchiectatic spaces, numerous coarse crackles may be heard over the affected areas.
2. Collapse with retained secretions blocking a proximal bronchus may lead to locally diminished breath sounds, with advanced disease may lead to scarring and overlying bronchial breathing.
3. Acute haemoptysis is an important complication of bronchiectasis

Investigations

- A. Bacteriological and mycological examination of sputum

1. In addition to common respiratory pathogens, sputum culture may reveal *Pseudomonas aeruginosa*, fungi such as *Aspergillus* and various mycobacteria.

2. Frequent cultures are necessary to ensure appropriate treatment of resistant organisms.

B. Radiological examination

1. Bronchiectasis, unless very gross, is not usually apparent on a chest X-ray.
2. In advanced disease, thickened airway walls, cystic bronchiectatic spaces, and associated areas of pneumonic consolidation or collapse may be visible.

C. CT

3. CT is much more sensitive, and shows thickened dilated airways (Fig. 19.28).

D. Assessment of ciliary function

Bronchial Asthma

The chronic airway inflammation, airflow obstruction and increased airway hyper-responsiveness leads to wheeze, cough, chest tightness and dyspnoea

Epidemiology

1. Current estimates suggest that asthma affects 300 million people world-wide and an additional 100 million persons will be diagnosed by 2025.
2. The socio-economic impact is enormous, particularly when poor control leads to days lost from school or work, unscheduled health-care visits and hospital admissions.

Etiology

1. Although the development, course of disease and response to treatment are influenced by genetic determinants
2. The rapid rise in the prevalence of asthma implies that environmental factors are critically important in terms of its expression.
3. The relationship between atopy (a propensity to produce IgE) and asthma is well established
4. Inhalation of an allergen into the airway include house dust mites, pets such as cats and dogs, pests such as cockroaches, and fungi

(particularly *Aspergillus*) allergic bronchopulmonary aspergillosis,

5. Allergic mechanism are also implicated in some cases of occupational asthma

6. Although the aetiology of asthma is often elusive, an attempt should be made to identify the agents that may contribute to the appearance or aggravation of the condition.

7. With regard to potential allergens, particular enquiry should be made into exposure to a pet cat, guinea pig, rabbit or horse, pest infestation, or mould growth following water damage to a home or building

Pathophysiology

Chronic airway inflammation, airflow obstruction and increased airway hyper-responsiveness are the key components in the pathogenesis of asthma.



Airway hyper-reactivity (AHR)-the tendency for airways to contract too easily and too much in response to triggers that have little or no effect in normal individuals



Other factors likely to be important include the degree of airway narrowing and the influence of neurogenic mechanisms



With increasing severity and chronicity of the disease, remodeling of the airway occurs



Leading to fibrosis of the airway wall, fixed narrowing of the airway and a reduced response to bronchodilator medication



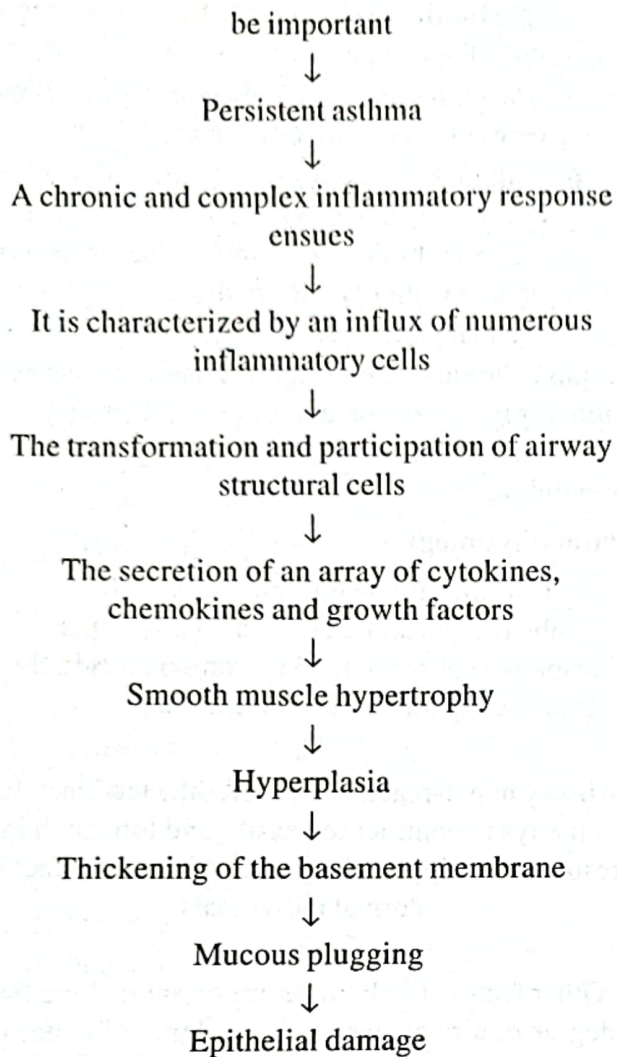
Exercise-induced asthma



Hyperventilation results in water loss from the pericellular lining fluid of the respiratory mucosa, which in turn triggers mediator release



Heat loss from the respiratory mucosa may also



Clinical features

1. Recurrent episodes of wheeze
2. Chest tightness
3. Breathlessness
4. Cough
5. Classical precipitants include exercise, particularly in cold weather, exposure to airborne allergens or pollutants, and viral upper respiratory tract infections.
6. Patients with mild intermittent asthma are usually asymptomatic between exacerbations.
7. Patients with persistent asthma report on-going breathlessness and wheeze, but variability is usually present with symptoms fluctuating over the course of one day, from day to day, or from month to month.
8. Asthma characteristically displays a diurnal

pattern, with symptoms and lung function being worse in the early morning.

9. In some circumstances, the appearance of asthma is triggered by medications. For example, β -adrenoreceptor antagonists (β -blockers), even when administered topically as eye drops, may induce bronchospasm, and aspirin and other non-steroidal anti-inflammatory drugs (NSAIDs) may also induce wheeze as above.

Diagnosis

1. The diagnosis of asthma is predominantly clinical and based on a characteristic history.
2. Supportive evidence is provided by the demonstration of variable airflow obstruction, preferably by using spirometry
3. The measurement of FEV1 and VC identify the obstructive nature of the ventilator defect, define its severity, and provide the basis of bronchodilator reversibility.
4. If spirometry is not available, a peak flow meter may be used.
5. Patients should be instructed to record peak flow readings after rising in the morning and before retiring in the evening.
6. A diurnal variation if PEF (the lowest values typically being recorded in the morning) of more than 20% is considered diagnostic and the magnitude of variability provides some indication of disease severity.
7. A trial of corticosteroids (e.g. 30 mg daily for 2 weeks) may be useful in documenting improvement in either FEV1 or PEF to establish the diagnosis.
8. When symptoms are predominantly related to exercise, an exercise challenge may be followed by a drop in lung function.
9. *Measurement of allergic status.* The presence of atopy may be demonstrated by skin prick tests.
10. The measurement of total and allergen-specific IgE.
11. A full blood picture may show peripheral blood eosinophilia.

12. *Radiological examination.* Chest X-ray appearances are often normal or show hyperinflation of lung fields. Lobar collapse may be seen if mucus occludes a large bronchus, and if accompanied by the presence of flitting infiltrates, may suggest that asthma has been complicated by allergic bronchopulmonary aspergillosis
13. An HRCT scan may be useful to detect bronchiectasis.

MCQs

- Persistent reduction in forced expiratory flow rates associated with increases in the residual volume and the residual volume/total lung capacity ratio, nonuniform distribution of ventilation, and ventilation-perfusion mismatching is the most typical finding in
 - COPD
 - BA
 - Lobar pneumonia
 - Interstitial lung disease
- Cough, weight loss, malaise, and gradual increase in dyspnea and inspiratory crackles and digital clubbing indicates
 - Hypersensitivity Pneumonitis
 - Tuberculosis
 - Chronic obstructive pulmonary disease (COPD)
 - Lung cancer
- Patients with severe airflow obstruction may also exhibit use of accessory muscles of respiration, sitting in the characteristic tripod position to facilitate the actions of the sternocleidomastoid, scalene, and intercostal muscles. Patients may develop cyanosis, visible in the lips and nail beds indicates
 - Bronchial Asthma
 - Chronic Obstructive Pulmonary Disease
 - Interstitial Lung Disease
 - Pneumonia
- Which among the following are obstructive lung disease
 - Bronchial asthma
 - Bronchiectasis
 - Bronchiolitis
 - Pulmonary sarcoidosis
 - I, II & III
 - I, II & IV
 - I, III&IV
 - II, III& IV

Answer

1.	a	2.	a	3.	b
4.	a				